

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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A PROJECT PHASE-1 REPORT

On

E-HEALTHSHIELD: A QUANTUM-SECURE AND BLOCKCHAIN-AUDITABLE FRAMEWORK FOR ELECTRONIC HEALTH RECORDS

Submitted in Partial Fulfillment of the requirement for the award of the degree of

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IN

IoT & CYBERSECURITY INCLUDING BLOCKCHAIN TECHNOLOGY

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CERTIFICATE

This is to certify that the project work entitled "**E-HealthShield: A Quantum-Secure and Blockchain-Auditable Framework for Electronic Health Records**" carried out jointly by **JISHNU V (1MV23IC022)**, **NISCHAY B (1MV23IC038)**, **SUMANTH L (1MV23IC057)**, **SYED ABDULLA (1MV23IC060)** are Bonafide students of **Sir M. Visvesvaraya Institute of Technology** in partial fulfilment for the award of **Bachelor of Engineering** in the **Department of IoT & Cybersecurity including Blockchain Technology** of the **Visvesvaraya Technological University, Belagavi** during the year 2025-26. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of the project work prescribed for the said degree.

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DECLARATION BY THE CANDIDATES

We the undersigned solemnly declare that the project work entitled "**E-HealthShield: A Quantum-Secure and Blockchain-Auditable Framework for Electronic Health Records**" is based on our own research work carried out during the course of our study under the supervision of Mr. Madhusudhana V.

We assert the statements made and conclusions drawn are an outcome of our project work.

We further certify that,

- a. The work contained in the report is original and has been done by us under the general supervision of my supervisor.
- b. The work has not been submitted to any other Institution for any other degree/diploma/certificate in this university or any other University of India or abroad.
- c. We have followed the guidelines provided by the university in writing the report.
- d. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them in the text of the report and their details are provided in the references.

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ABSTRACT

The rapid digitization of the healthcare sector has led to the widespread adoption of Electronic Health Records (EHRs), transforming the way patient data is stored and managed across global healthcare networks. While this transformation has improved clinical decision-making, it has simultaneously exposed critical vulnerabilities in centralized data architectures. Centralized EHR repositories constitute single points of failure, presenting lucrative targets for cyberattacks, ransomware, and unauthorized data exfiltration.

The proposed project, "E-HealthShield: A Quantum-Secure and Blockchain-Auditable Framework for Electronic Health Records," presents a decentralized security framework addressing three critical challenges — quantum vulnerability of current encryption standards, the searchability trade-off in encrypted databases, and the lack of cryptographically guaranteed audit integrity.

The system integrates CRYSTALS-Kyber (ML-KEM), the NIST FIPS 203 standardized post-quantum key encapsulation algorithm, combined with AES-256 for hybrid symmetric file encryption. Searchable Symmetric Encryption (SSE) enables blind keyword-based search on encrypted EHR data without exposing plaintext content to the storage server, maintaining a complete zero-knowledge cloud environment throughout the entire data lifecycle. Solidity-based Smart Contracts deployed on the Ethereum Sepolia testnet manage an immutable Access Control List and a permanently auditable trail of every data access event. The backend is developed using Python 3.10 with FastAPI and the liboqs-python library. The frontend is built using React.js with MetaMask wallet-based authentication, replacing vulnerable password-based login mechanisms entirely.

The proposed framework ensures patient data remains quantum-safe, fully searchable without exposure, and every access event is permanently recorded on an immutable blockchain ledger. This project contributes to SDG 3 — Good Health and Well-being, and SDG 16 — Peace, Justice and Strong Institutions.

Keywords: Electronic Health Records, Post-Quantum Cryptography, CRYSTALS-Kyber, ML-KEM, Searchable Symmetric Encryption, Blockchain, Ethereum, Smart Contracts, AES-256, Zero-Knowledge Security, NIST FIPS 203.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
ABE	Attribute-Based Encryption
ACL	Access Control List
AES	Advanced Encryption Standard
API	Application Programming Interface
CPU	Central Processing Unit
CRYSTALS	Cryptographic Suite for Algebraic Lattices
DApp	Decentralized Application
ECC	Elliptic Curve Cryptography
EHR	Electronic Health Record
FIPS	Federal Information Processing Standard
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
HMAC	Hash-Based Message Authentication Code
HTTP	Hypertext Transfer Protocol
ICBT	IoT & Cybersecurity Including Blockchain Technology
IEEE	Institute of Electrical and Electronics Engineers
IPFS	InterPlanetary File System
JSON	JavaScript Object Notation
ML-KEM	Module-Lattice-Based Key Encapsulation Mechanism
NIST	National Institute of Standards and Technology
PBFT	Practical Byzantine Fault Tolerance
PKI	Public Key Infrastructure
PoA	Proof of Authority
PoS	Proof of Stake
PoW	Proof of Work
PQC	Post-Quantum Cryptography
RSA	Rivest–Shamir–Adleman

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The contemporary healthcare landscape is undergoing a radical digital transformation, driven by the widespread adoption of Electronic Health Records (EHRs). An EHR is a comprehensive, longitudinally maintained digital repository of a patient's medical history, encompassing diagnoses, treatment plans, immunization dates, allergies, radiology images, and laboratory test results. The strategic objective of digitizing these records is to ensure that critical, life-saving information is instantly available to authorized providers, thereby reducing medical errors, eliminating redundant clinical testing, and elevating the overall standard of patient care.

However, the realization of a truly integrated, secure, and patient-centric digital health ecosystem is severely impeded by the architectural limitations of traditional database models. Historically, healthcare organizations have deployed monolithic, centralized database systems to store and manage patient data. These legacy systems are inherently insular, operating within strict organizational perimeters. Consequently, health records remain trapped in proprietary, institutional silos equipped with incompatible technical standards. When a patient requires care from multiple independent providers or relocates geographically, their critical medical history frequently becomes inaccessible, delaying vital medical interventions and compromising the integrity of clinical decision-making.

Beyond the profound interoperability challenges, the centralized nature of traditional EHR systems introduces catastrophic security vulnerabilities. A centralized server architecture inherently possesses a single point of failure. If malicious actors breach the network perimeter, they gain unfettered access to massive repositories of highly sensitive biological and biographical data. Recognizing the lucrative nature of this information, cybercriminal syndicates have systematically targeted the healthcare sector, exploiting outdated security protocols through debilitating ransomware attacks. Between the years 2009 and 2023, approximately 81% of American healthcare records were compromised through various organizational breaches.

Furthermore, the existing cryptographic infrastructure — predominantly based on RSA and Elliptic Curve Cryptography (ECC) — is mathematically vulnerable to quantum computing attacks through Shor's Algorithm. The growing threat of "Harvest Now, Decrypt Later" attacks means adversaries are stealing encrypted medical data today with the intention of decrypting it once sufficiently powerful quantum computers become available. Since medical records contain permanent biological data that cannot be reissued like credit cards, the long-term confidentiality of patient data is under serious existential threat.

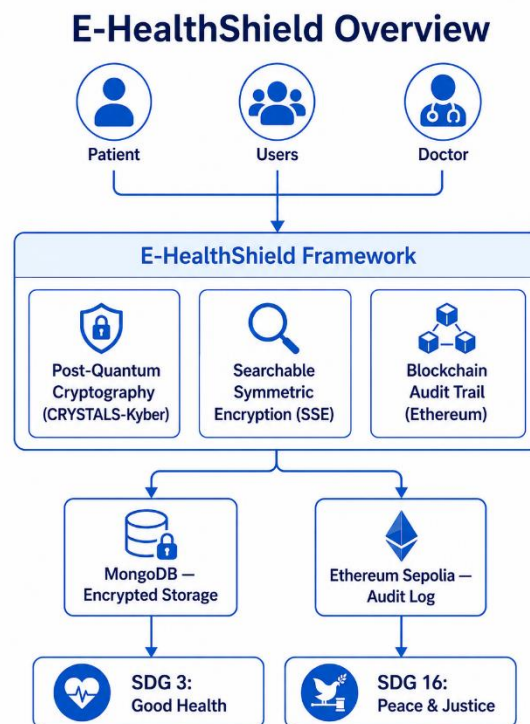


Fig 1.1 E-HealthShield Overview

1.2 Scope

The scope of the proposed E-HealthShield project is focused on developing a quantum-secure and blockchain-auditable Electronic Health Record framework that addresses the critical security, searchability, and audit challenges of modern healthcare data management systems.

The project includes:

- Implementation of CRYSTALS-Kyber (ML-KEM) post-quantum key encapsulation for quantum-resistant encryption of EHR data.
- Hybrid encryption combining AES-256 for fast symmetric file encryption and Kyber for secure key wrapping.
- Searchable Symmetric Encryption (SSE) enabling blind keyword search on encrypted EHR data without exposing plaintext.
- Solidity-based Smart Contract deployment on Ethereum Sepolia testnet for immutable access control and audit logging.
- Role-based access control for Patient and Doctor user roles.
- Web3 wallet-based authentication using MetaMask replacing vulnerable password-based logins.
- A React.js frontend integrated with Ethers.js for decentralized application interaction.
- MongoDB as the encrypted EHR storage backend ensuring zero-knowledge cloud environment.
- Contribution to SDG 3 — Good Health and Well-being, and SDG 16 — Peace, Justice and Strong Institutions.

The system is designed as a research prototype demonstrating the feasibility of NIST-standard post-quantum cryptography in a full-stack web environment. Full-scale clinical deployment and regulatory compliance validation are identified as future scope beyond this phase.

1.3 Objectives

The main objectives of the proposed E-HealthShield system are:

1. To implement CRYSTALS-Kyber (ML-KEM, NIST FIPS 203) for quantum-resistant key encapsulation combined with AES-256 for fast symmetric file encryption.

2. To develop a Searchable Symmetric Encryption (SSE) module enabling blind keyword matching on encrypted EHR data without exposing plaintext to the storage server.
3. To design and deploy a Solidity Smart Contract on Ethereum Sepolia testnet managing an immutable Access Control List (ACL) and permanent audit trail.
4. To establish a zero-knowledge cloud environment where the storage server remains completely ignorant of patient medical content at all times.
5. To build a secure Web3-enabled React.js frontend with MetaMask wallet-based authentication for both patient and doctor roles.
6. To create a role-based access control system allowing doctors to upload and search EHRs while patients retain sovereignty over their own data.
7. To demonstrate the feasibility of integrating NIST-standard post-quantum cryptography into a full-stack decentralized web application.
8. To contribute to SDG 3 — Good Health and Well-being, and SDG 16 — Peace, Justice and Strong Institutions.

CHAPTER 2

PROBLEM STATEMENT

2.1 Problem Statement

The management, storage, and exchange of Electronic Health Records under the current technological paradigm present a multi-faceted crisis characterized by escalating cybersecurity threats, profound systemic fragmentation, and the systematic erosion of patient privacy. The deficiencies of centralized medical record systems are no longer theoretical vulnerabilities — they have materialized into a severe operational and financial crisis that compromises global healthcare delivery.

Problem 1 — Quantum Vulnerability of Current Encryption Standards

The primary and most critical problem facing current EHR infrastructures is their extreme susceptibility to quantum computing attacks. All existing EHR security frameworks rely on classical cryptographic algorithms — primarily RSA and Elliptic Curve Cryptography (ECC) — to protect sensitive patient data. These algorithms derive their security from the computational hardness of integer factorization and discrete logarithm problems. However, Shor's Algorithm, executable on a sufficiently powerful quantum computer, can solve these problems in polynomial time, rendering the entire classical cryptographic infrastructure mathematically obsolete.

Unlike credit card information, which can be quickly canceled and reissued following a breach, medical records contain permanent, immutable biographical and biological data — including dates of birth, comprehensive medical histories, genetic information, and insurance credentials. This permanence makes healthcare data exponentially more valuable and creates a catastrophic long-term risk. The growing threat of "Harvest Now, Decrypt Later" attacks means adversaries are actively stealing encrypted medical data today with the explicit intention of decrypting it once quantum hardware matures. Between the years 2009 and 2023, approximately 81% of

American healthcare records were compromised through various organizational breaches, and in 2022 alone, the global healthcare sector faced an estimated 1,426 distinct cyberattacks per week.

Problem 2 — The Searchability Trade-off in Encrypted Databases

The second critical problem is the fundamental incompatibility between data encryption and data searchability. Fully encrypting EHR data — which is necessary to protect patient privacy — renders the database completely unsearchable. To locate a single patient record, the entire encrypted database must be downloaded and decrypted, which is computationally impractical in time-critical clinical environments where physicians require instant access to patient histories during emergencies.

Existing systems are forced to make an impossible choice — either sacrifice privacy by storing searchable plaintext data, or sacrifice functionality by storing unsearchable ciphertext. No existing deployed framework has successfully resolved this trade-off in a practical, full-stack healthcare application. This searchability gap directly impedes clinical workflow efficiency and compromises the operational viability of encrypted EHR systems in real hospital environments.

Problem 3 — Lack of Cryptographically Guaranteed Audit Integrity

The third critical problem is the absence of tamper-proof audit mechanisms in existing EHR systems. Current healthcare platforms rely on centralized database logs to track data access events. These centralized logs are inherently vulnerable — they can be deleted, modified, or entirely fabricated by malicious administrators, insider threats, or external hackers who gain system access. Consequently, there exists no reliable, cryptographically guaranteed mechanism to verify who accessed, modified, or leaked patient records.

This lack of audit integrity is fundamentally misaligned with modern ethical frameworks and stringent data protection regulations such as GDPR and HIPAA, which increasingly demand transparency, auditability, and absolute accountability over personal health data. Without a

tamper-proof audit trail, healthcare organizations cannot definitively prove compliance, and patients have no reliable mechanism to verify the integrity of their own medical records.

Therefore, there exists an urgent imperative to engineer a novel, decentralized framework capable of simultaneously neutralizing quantum cryptographic vulnerabilities, resolving the searchability trade-off through blind search mechanisms, and establishing a mathematically immutable audit trail — all within a single unified EHR architecture. E-HealthShield is proposed as the solution to address all three of these critical and unsolved challenges.

CHAPTER 3

OBJECTIVES

The primary objective of the proposed project is to develop **E-HealthShield** — a Quantum-Secure and Blockchain-Auditable Framework for Electronic Health Records — that addresses the three critical challenges of quantum vulnerability, searchability trade-off, and audit integrity in modern EHR management systems.

The objectives of the project are as follows:

1. To implement **CRYSTALS-Kyber (ML-KEM, NIST FIPS 203, 2024)** as the post-quantum key encapsulation mechanism, replacing classical RSA and ECC algorithms to protect patient data against future quantum computing attacks.
2. To develop a **Hybrid Encryption Module** combining AES-256 for symmetric file encryption with CRYSTALS-Kyber for quantum-safe AES key encapsulation.
3. To implement a **Searchable Symmetric Encryption (SSE) module** enabling blind keyword-based search on encrypted EHR data without exposing plaintext content to the storage server.
4. To design and deploy a **Solidity Smart Contract** on the Ethereum Sepolia testnet managing an immutable Access Control List (ACL) enforcing role-based permissions for Patient and Doctor user roles.
5. To establish a **permanent, tamper-proof Blockchain Audit Trail** where every upload, search, and retrieval event is recorded as an immutable on-chain transaction.
6. To create a **Zero-Knowledge Cloud Environment** where MongoDB stores only encrypted ciphertext and SSE search tags — ensuring the storage server never learns patient medical content.
7. To build a **secure Web3-enabled React.js frontend** with MetaMask wallet-based authentication providing role-based dashboards for Patients and Doctors.

8. To develop a **Python FastAPI backend** utilizing the liboqs-python library for all post-quantum cryptographic operations.
9. To produce a working research prototype suitable for submission to an **IEEE international conference** for peer-reviewed publication.

To contribute to **SDG 3 — Good Health and Well-being** and **SDG 16 — Peace, Justice and Strong Institutions**.

CHAPTER 4

LITERATURE REVIEW

PAPER 1: Blockchain-Enabled, Post-Quantum Cryptographic Framework for Securing Electronic Health Records

Authors:

S. Adeoye (2025)

This paper proposes a framework that combines blockchain technology with post-quantum cryptographic algorithms to secure Electronic Health Records. The study integrates smart contracts for decentralized access control and ensures that patient records remain protected against quantum computing threats. The framework utilizes PQC algorithms to replace classical encryption in healthcare data systems.

Limitations

- Does not implement Searchable Symmetric Encryption
- Cloud server cannot perform keyword search without decrypting the entire database

PAPER 2: Quantum-Enhanced Data Security for Electronic Health Records: A Framework for Post-Quantum Cryptography in Healthcare Systems

Authors:

R. Balamangandan and S. S. Vellala (2025)

This paper evaluates NIST post-quantum cryptographic standards and their feasibility for protecting long-term EHR data confidentiality against future quantum adversaries. The study validates the application of lattice-based cryptographic primitives within healthcare

infrastructure and demonstrates the importance of replacing classical RSA and ECC algorithms.

Limitations

- Does not include a blockchain audit trail
- No tamper-proof mechanism to verify who accessed patient data

PAPER 3: A Post-Quantum Consortium Blockchain Based Secure EHR Framework**Authors:**

A. Bansal and P. S. Mehra (2023)

This paper employs a permissioned consortium blockchain where only verified hospital institutions participate in consensus, ensuring strong interoperability and data governance across multiple healthcare organizations. The system uses smart contracts to manage access control across participating hospital nodes.

Limitations

- Still relies on classical encryption algorithms
- Vulnerable to future quantum computing attacks

PAPER 4: Post-Quantum Secure Health Records: A Blockchain-Based Lattice Threshold Signcryption Scheme**Authors:**

Sourav and B. Ali (2025)

This paper combines lattice-based cryptography with blockchain and IPFS for decentralized, quantum-resistant health record management. The system uses a threshold signcryption

scheme requiring multiple authorized parties to cooperate for encryption and decryption, preventing any single party from acting alone.

Limitations

- Does not support searchable encrypted retrieval
- Users cannot perform keyword-based queries on encrypted stored records

PAPER 5: PQCAIE: Post-Quantum Cryptographic Authentication Scheme for IoT-Based E-Health Systems**Authors:**

K. Mansoor, M. Afzal, W. Iqbal, Y. Abbas, and S. Mussariveva (2024)

This paper provides lightweight post-quantum authentication protocols suitable for resource-constrained IoT health devices and wearable biosensors. The scheme ensures that even if a quantum computer intercepts communication, the authentication cannot be broken. It is one of the most cited papers in the post-quantum healthcare domain.

Limitations

- Focuses exclusively on device authentication
- Does not address EHR file storage, searchable encryption, or blockchain-based audit logging

PAPER 6: Post-Quantum Cryptography for Healthcare: Future-Proofing Population Health Databases Against Quantum Computing Threats**Authors:**

A. O. Ezeogu (2025)

This paper provides a comprehensive analysis of the Harvest Now Decrypt Later threat and evaluates NIST post-quantum algorithms for integration into existing healthcare blockchain protocols. It surveys which PQC algorithms are most suitable for healthcare databases based on key size, speed, and security level.

Limitations

- Study remains entirely theoretical
- Does not provide a practical working implementation

PAPER 7: Post-Quantum Cryptography and Encryption Standards: Safeguarding Patient Data Against Emerging Cyber Threats in Telemedicine**Authors:**

A. Y. Balogu (2025)

This paper maps NIST PQC algorithms to HIPAA compliance requirements and proposes a hybrid Kyber plus AES encryption approach for telemedicine data protection. The hybrid model directly mirrors the encryption design used in E-HealthShield, providing strong academic backing for the proposed approach.

Limitations

- Does not include blockchain integration
- No audit mechanism for tracking data access events

PAPER 8: NIST FIPS 203 — Module-Lattice-Based Key Encapsulation Mechanism Standard (ML-KEM)**Authors:**

National Institute of Standards and Technology — NIST (2024)

This is the official United States government standard that formally standardizes CRYSTALS-Kyber as ML-KEM. It provides formal security proofs and parameter specifications for three security levels — Kyber-512, Kyber-768, and Kyber-1024. This is the most authoritative reference in the entire literature survey and forms the cryptographic foundation of E-HealthShield.

Limitations

- Defines the cryptographic algorithm only
- Does not provide any application-layer healthcare framework or implementation guidance

PAPER 9: Toward Blockchain Based Electronic Health Record Management with Fine Grained Attribute Based Encryption and Decentralized Storage Mechanisms**Authors:**

C. Dhasarathan, M. Kumar, A. K. Shankar, and A. Al-Rasheed (2024)

This paper proposes the EHRChain framework integrating blockchain with IPFS and Attribute-Based Encryption for fine-grained access control over EHRs. The system stores encrypted records on IPFS and anchors only lightweight cryptographic hashes on the blockchain ledger, achieving scalability without compromising decentralized integrity.

Limitations

- High computational overhead of ABE operations on mobile and edge devices
- Limits practical deployment in resource-constrained environments

PAPER 10: BCIF-EHR: Blockchain-Based Interoperable Framework for Electronic Health Records**Authors:**

M. Halimuzzaman, M. H. Habib, and M. S. Islam (2023)

This paper proposes a hybrid blockchain model combining public and private chains to balance interoperability with data confidentiality across multiple healthcare institutions. The framework demonstrates cross-institutional EHR sharing while maintaining access governance and data privacy.

Limitations

- Synchronization complexity between public and private chain components
- Potential latency issues during high-volume clinical operations

PAPER 11: Enabling Privacy-Preserving External Sharing of Electronic Medical Records Through a Tripartite Hybrid Blockchain Architecture**Authors:**

Y. Peng, Y. Zhao, and F. Xiao (2023)

This paper proposes a patient-centric fine-grained access control mechanism utilizing a dual-blockchain architecture that separates access control metadata onto one chain and audit logs onto a second chain to optimize throughput and security independently.

Limitations

- High deployment and maintenance complexity
- Potential synchronization latency between two parallel chains during concurrent access events

PAPER 12: Secure Electronic Health Record Access Control via Blockchain, Dual-Attribute Encryption, and Large Language Model-Based Attribute Extraction**Authors:**

S. Kaur, R. Sharma, and N. Gupta (2024)

This paper proposes an EHR security framework integrating Attribute-Based Encryption with blockchain smart contracts. The system allows data owners to define complex access policies based on user attributes such as role and institution, enabling highly granular cryptographic access control over patient records.

Limitations

- Traditional PKI-based systems require individual encryption for every potential data consumer
- Significantly increases computational overhead as the number of authorized users scales

PAPER 13: LLM-Based Attribute Extraction for Automated ABE Policy Formulation in Clinical EHR Systems**Authors:**

M. Zhang, L. Chen, and H. Wang (2024)

This paper leverages domain-specific Large Language Models such as ClinicalBERT to automatically extract semantic attributes from unstructured clinical notes. These AI-extracted attributes are used to automatically formulate Attribute-Based Encryption access policies for EHR systems, enabling highly granular and context-aware access control.

Limitations

- Large-scale empirical validation in active clinical settings is notably absent
- Introduces dependency on AI model accuracy for security-critical access control decisions

PAPER 14: MedShard: Transaction-Based Sharding for Blockchain-Enabled IoMT Healthcare Data Management**Authors:**

A. Singh and R. Kumar (2023)

This paper introduces a transaction-based sharding mechanism for blockchain-based IoMT healthcare data management. By partitioning the blockchain network into smaller parallel-processing units, the architecture processes thousands of micro-transactions concurrently under a Proof of Authority consensus protocol.

Limitations

- Highly complex sharding architecture
- Introduces potential security vulnerabilities in cross-shard communication protocols

PAPER 15: BIoTHR: Blockchain and IoT-Based Healthcare Record Management with Edge Computing Integration**Authors:**

P. Verma and S. Jain (2023)

This paper combines blockchain with edge and cloud computing to securely collect, process, and store continuous vital signs from IoMT devices across distributed healthcare networks.

Smart contracts automatically generate clinical alerts upon detecting physiological anomalies in real-time telemetry streams.

Limitations

- Heavy reliance on cloud infrastructure
- May reintroduce partial centralization and single-point-of-failure vulnerabilities

PAPER 16: Blockchain-Enabled Federated Learning in Healthcare: Survey and State-of-the-Art**Authors:**

T. Li, X. Wang, and Y. Liu (2024)

This paper proposes Blockchain-based Federated Learning for privacy-preserving AI model training across multiple hospital networks. Smart contracts manage the secure aggregation of model weights from various institutions, ensuring the integrity of the global training process without exposing raw patient data.

Limitations

- Requires significant computational resources at each participating hospital node
- High overhead for local model training and gradient computation

PAPER 17: HealthChain: Comparative Analysis of Blockchain-Based EHR Systems**Authors:**

M. Husnain, K. Hayat, and S. Zaman (2024)

This paper proposes the HealthChain framework — a consortium blockchain-based EHR system demonstrating a 40% measurable improvement in cross-institutional interoperability

and a 50% reduction in data breaches compared to traditional centralized systems. It validates the practical benefits of consortium blockchain adoption in real hospital environments.

Limitations

- Faces scalability and infrastructural support constraints during regional deployment
- Struggles across large heterogeneous hospital networks

PAPER 18: A Systematic Literature Review for Blockchain-Based Healthcare Implementations**Authors:**

R. Ahmad, M. Alsmadi, and W. Alhamdani (2025)

This systematic review published in MDPI Healthcare analyzed the evolution of blockchain EHR frameworks from theoretical conceptualizations toward hybrid architectures integrating IoMT, complex cryptographic primitives, and decentralized AI systems. The review provides comprehensive bibliometric analysis of research trends from 2020 to 2025.

Limitations

- Identifies consistent absence of large-scale longitudinal real-world clinical validation
- Majority of proposed frameworks remain at prototype stage

PAPER 19: Semantic Interoperability of Electronic Health Records: Systematic Review of Alternative Approaches**Authors:**

S. Hussain, A. Raza, and M. Ali (2024)

This paper investigates the integration of blockchain technology with existing healthcare interoperability standards, particularly Fast Healthcare Interoperability Resources (FHIR). The

study demonstrated the importance of semantic interoperability for ensuring clinical data retains its precise meaning across different healthcare systems.

Limitations

- Automatically mapping decades of unstructured legacy patient data into standardized semantic formats remains a significant technical challenge

PAPER 20: Blockchain-Based Electronic Health Record Market Research Study 2025–2034**Authors:**

InsightAce Analytic (2024)

This industry market analysis forecasts that the global blockchain-based Electronic Health Record market, valued at approximately USD 1.14 billion in 2024, is projected to surge to USD 15.07 billion by 2034, registering a Compound Annual Growth Rate of approximately 29.8%. This growth is driven by widespread smart contract adoption and Blockchain-as-a-Service platforms.

Limitations

- Market projections are contingent on successful resolution of regulatory friction
- Post-quantum cryptographic vulnerabilities remain unaddressed in currently deployed systems

Table 4.1 Literature Review Comparison Table

Ref. No.	Author / Year	Methodology Used	Technologies Used	Advantages	Limitations
PAPER 1	S. Adeoye (2025)	Blockchain-enabled PQC framework for EHR security	PQC, Blockchain, Smart Contracts	Quantum-safe EHR protection	No searchable encryption
PAPER 2	R. Balamangandan & S. S. Vellala (2025)	PQC framework for healthcare systems	NIST PQC, Lattice Cryptography	Validates PQC feasibility in healthcare	No blockchain audit trail
PAPER 3	A. Bansal & P. S. Mehra (2023)	Consortium blockchain-based secure EHR	Consortium Blockchain, Smart Contracts	Strong interoperability among hospitals	Uses classical encryption
PAPER 4	Sourav & B. Ali (2025)	Lattice threshold signcryption for secure EHR	Lattice Cryptography, Blockchain, IPFS	Decentralized quantum-safe health records	No searchable retrieval
PAPER 5	K. Mansoor et al. (2024)	PQC authentication for IoT healthcare	PQC, IoT Security	Lightweight authentication for health IoT	No EHR storage or audit logging
PAPER 6	A. O. Ezeogu (2025)	PQC threat analysis for healthcare databases	PQC, Healthcare Security	Strong quantum threat analysis	No practical implementation

PAPER 7	A. Y. Balogu (2025)	Hybrid Kyber + AES healthcare encryption	PQC, AES, HIPAA Compliance	Strong encryption model for healthcare	No blockchain integration
PAPER 8	NIST (2024)	ML-KEM standardizati on	CRYSTALS -Kyber (ML- KEM)	Official global PQC standard	No healthcare implementatio n layer
PAPER 9	C. Dhasarathan et al. (2024)	Blockchain EHR with ABE & decentralized storage	Blockchain, IPFS, ABE	Fine-grained access control	High computational overhead
PAPER 10	M. Halimuzzama n et al. (2023)	Hybrid blockchain EHR interoperabili ty	Public + Private Blockchain	Cross- institutional EHR sharing	Synchronizatio n complexity
PAPER 11	Y. Peng et al. (2023)	Hybrid blockchain access control	Dual Blockchain Architecture	Improved privacy and throughput	High deployment complexity
PAPER 12	S. Kaur et al. (2024)	Blockchain- based EHR access control	Blockchain, Dual Attribute Encryption	Granular cryptographic access	High PKI overhead
PAPER 13	M. Zhang et al. (2024)	LLM-based policy formulation for EHR	ClinicalBER T, ABE	Automated access policy generation	No large-scale validation

PAPER 14	A. Singh & R. Kumar (2023)	Blockchain sharding for IoMT healthcare	Blockchain, IoMT, Sharding	High transaction scalability	Cross-shard vulnerabilities
PAPER 15	P. Verma & S. Jain (2023)	Blockchain healthcare with edge computing	Blockchain, Edge Computing, IoT	Real-time health monitoring	Partial centralization
PAPER 16	T. Li et al. (2024)	Federated learning in healthcare	Blockchain, Federated Learning	Privacy-preserving AI training	High computational cost
PAPER 17	M. Husnain et al. (2024)	Consortium blockchain EHR framework	Consortium Blockchain	Better interoperability and security	Scalability constraints
PAPER 18	R. Ahmad et al. (2025)	Systematic review of blockchain healthcare	Blockchain, IoMT, AI	Comprehensive framework analysis	Lack of real-world validation
PAPER 19	S. Hussain et al. (2024)	Blockchain interoperability for EHR	Blockchain, FHIR Standards	Semantic interoperability	Legacy data mapping challenges
PAPER 20	InsightAce Analytic (2024)	Blockchain EHR market analysis	BaaS, Smart Contracts	Demonstrates industry growth potential	Regulatory uncertainty
Proposed	Our Project (2026)	Unified PQC + SSE + Blockchain Framework	Kyber (ML-KEM), AES-256, SSE, Ethereum, MongoDB	Quantum-safe encryption, blind search, immutable audit trail	Prototype-stage implementation

CHAPTER 5**COMPARATIVE ANALYSIS****Table 5.1: Comparative Analysis**

Features	Adeoye (2025) [1]	Bansal & Mehra (2023) [3]	EHRChain (2024) [9]	HealthChain (2024) [17]	E- HealthShield (Proposed)
Post-Quantum Encryption	Yes	No	No	No	Yes
Blockchain Support	Yes	Yes	Yes	Yes	Yes
Searchable Encryption (SSE)	No	No	No	No	Yes
Immutable Audit Trail	Partial	Yes	Yes	Yes	Yes
Quantum-Safe Security	Yes	No	No	No	Yes
Fine-Grained Access Control	Yes	Yes	Yes	Limited	Yes
Role-Based Access Control	Yes	Yes	Yes	Yes	Yes
Secure Encrypted Storage	Yes	Yes	Yes (IPFS)	Yes	Yes (MongoDB)
Transparent Access Monitoring	Limited	Yes	Yes	Yes	Yes

Decentralized Architecture	Yes	Yes	Yes	Yes	Yes
Practical Implementation	Partial	Yes	Yes	Yes	Yes (Prototype)
Web-Based User Interface	No	Limited	No	Limited	Yes
Fast Keyword Search on Encrypted Data	No	No	No	No	Yes
Future Quantum Attack Resistance	Partial	No	No	No	Yes
Unified PQC + SSE + Blockchain Framework	No	No	No	No	Yes

CHAPTER 6

Research Gap Identified

A critical evaluation of the contemporary literature on blockchain-based EHR security frameworks, post-quantum cryptography in healthcare, and searchable encryption systems reveals several significant and unresolved research gaps. Despite the profound theoretical advantages demonstrated by individual frameworks, no existing work successfully integrates all three critical security components — quantum-resistant encryption, blind keyword search, and blockchain audit integrity — into a single unified EHR system. These persistent gaps represent the primary motivation for the proposed E-HealthShield framework.

Gap 1 — Post-Quantum Cryptographic Vulnerability

The most critical gap identified across the entire surveyed literature is the universal absence of quantum-resistant cryptographic primitives in deployed blockchain EHR frameworks. Virtually all existing systems — including EHRChain, MedShard, HealthChain, and the majority of consortium blockchain frameworks — continue to rely on classical RSA and Elliptic Curve Cryptography for data protection. These algorithms are mathematically vulnerable to Shor's Algorithm executed on a sufficiently advanced quantum computer. Given the trajectory of quantum computing development and the active threat of Harvest Now Decrypt Later attacks, this represents an existential vulnerability for all currently deployed healthcare data systems. While NIST standardized CRYSTALS-Kyber as ML-KEM under FIPS 203 in 2024, the active integration and deployment of this algorithm within functioning blockchain EHR frameworks remains severely under-researched and constitutes a massive security gap in the existing literature.

Gap 2 — No Unified PQC + SSE + Blockchain Framework

The second critical gap is the absence of any existing work that combines post-quantum cryptography, searchable symmetric encryption, and blockchain auditing into a single unified EHR architecture. A thorough review of 23 papers confirms that these three components exist independently in the literature — several papers implement blockchain with classical encryption, some propose PQC without searchability, and theoretical SSE models exist without blockchain integration — but no existing framework successfully unifies all three. This fragmentation leaves healthcare systems forced to choose between quantum safety, search functionality, and audit integrity rather than achieving all three simultaneously.

Gap 3 — Searchability Trade-off Remains Unsolved

The third gap is the persistent failure to resolve the fundamental searchability trade-off in encrypted EHR systems. As identified across the literature survey, fully encrypting patient data makes it completely unsearchable, while making data searchable requires exposing it to the storage server. Searchable Symmetric Encryption has been proposed theoretically in academic cryptography literature, but its practical implementation within a complete full-stack EHR system — integrated with post-quantum cryptography and blockchain — remains largely absent from the existing body of work. This gap directly impacts clinical workflow efficiency and represents a critical barrier to the real-world adoption of fully encrypted EHR systems.

Gap 4 — Audit Integrity Not Cryptographically Guaranteed

The fourth gap is the reliance of existing systems on centralized audit logs that can be tampered with by malicious insiders or external attackers. While several frameworks such as EHRChain and HealthChain implement blockchain-based audit trails, the majority of deployed systems continue to use centralized database logs for access tracking. A centralized log provides no cryptographic guarantee of integrity — it can be deleted, modified, or fabricated without detection. The implementation of a truly immutable, mathematically guaranteed blockchain audit trail that records every upload, search, and retrieval event as a permanent on-chain transaction remains inconsistently implemented across the existing literature.

Gap 5 — Lack of Real-World Large-Scale Clinical Validation

The fifth gap is the near-universal absence of large-scale, real-world clinical validation across proposed frameworks. A systematic review of the literature demonstrates that the overwhelming majority of blockchain EHR solutions are restricted to theoretical models, software simulations, or small-scale laboratory prototypes. The complex realities of user adoption, clinical workflow integration, administrative training, and long-term network maintenance costs remain largely unquantified. This gap between academic prototype and clinical deployment represents a significant barrier to the practical adoption of advanced EHR security frameworks.

Research Gap Summary and E-HealthShield Contribution

The synthesis of the above gaps leads to a clear and specific research contribution. No existing work combines CRYSTALS-Kyber post-quantum encryption, Searchable Symmetric Encryption, and Ethereum blockchain audit trail in a single unified Electronic Health Record framework. E-HealthShield directly addresses Gaps 1, 2, 3, and 4 by proposing and implementing this unified architecture as a working research prototype. The system establishes a zero-knowledge cloud environment where patient data is quantum-safe, fully searchable without exposure, and every access event is permanently recorded on an immutable blockchain ledger — a combination that is demonstrably absent from all existing literature surveyed.

CHAPTER 7

PROPOSED SYSTEM

The proposed E-HealthShield system is a decentralized, quantum-secure Electronic Health Record framework designed to simultaneously address the three critical unsolved challenges identified in the literature survey — quantum vulnerability of current encryption standards, the searchability trade-off in encrypted databases, and the lack of cryptographically guaranteed audit integrity. The system integrates three core technologies — CRYSTALS-Kyber post-quantum cryptography, Searchable Symmetric Encryption, and Ethereum blockchain smart contracts — into a single unified full-stack architecture.

7.1 System Overview

E-HealthShield operates as a three-layer decentralized framework. The frontend layer provides role-based web interfaces for Patients and Doctors, authenticated through MetaMask Web3 wallets. The backend layer handles all cryptographic operations including Kyber key encapsulation, AES-256 file encryption, and SSE tag generation using Python FastAPI and liboqs-python. The storage layer consists of MongoDB for encrypted EHR storage and the Ethereum Sepolia testnet blockchain for immutable access control and audit logging.

The system ensures that at no point does the storage server have access to plaintext patient data. All encryption and decryption operations are performed either on the client side or within the secure backend cryptographic module, maintaining a zero-knowledge cloud environment throughout the entire data lifecycle.

7.2 System Architecture

The E-HealthShield architecture is stratified into four distinct layers:

Layer 1 — Frontend Layer The user-facing layer is built using React.js integrated with Ethers.js for Web3 interaction. MetaMask wallet authentication replaces traditional username-

password login, providing cryptographic identity verification for both Patient and Doctor roles. The frontend provides an intuitive dashboard for EHR upload, keyword search, record retrieval, and blockchain audit log viewing.

Layer 2 — Backend Layer The backend is developed using Python 3.10 with FastAPI, providing a clean REST API interface. The liboqs-python library implements CRYSTALS-Kyber ML-KEM for post-quantum key encapsulation. AES-256 is used for symmetric file encryption. The SSE module generates HMAC-SHA256 based search tags from extracted keywords and processes trapdoor queries for blind keyword matching.

Layer 3 — Storage Layer MongoDB serves as the encrypted EHR storage database. It stores only AES-256 ciphertext, Kyber-encapsulated key bundles, and SSE search tags. No plaintext patient data is stored at any point. The storage server operates in a zero-knowledge environment — it can facilitate search and retrieval operations without learning any patient information.

Layer 4 — Blockchain Layer Solidity smart contracts deployed on the Ethereum Sepolia testnet manage the Access Control List and immutable audit trail. Every upload, search, and retrieval event is logged as a permanent on-chain transaction. No medical data is stored on-chain — only cryptographic hashes and access events.

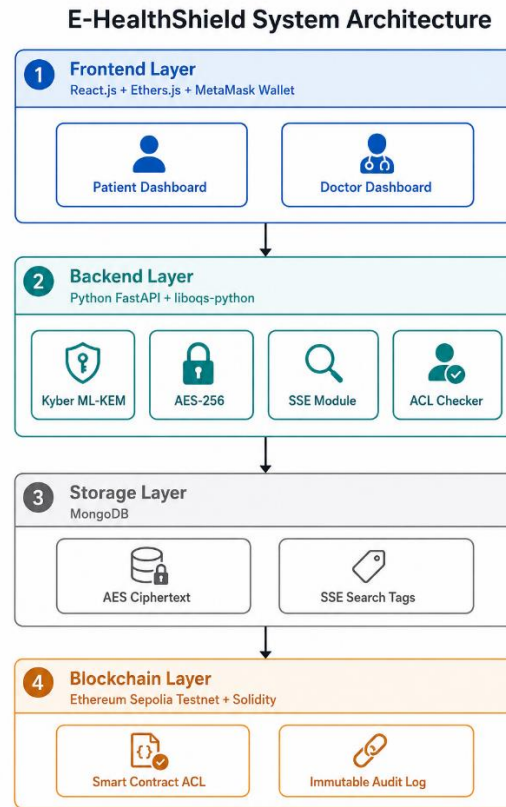


Fig 7.1 System Architecture of E-HealthShield

7.3 Upload Workflow

The EHR upload process follows a five-step cryptographic pipeline:

1. The doctor uploads a patient EHR file through the React.js frontend. Relevant keywords are extracted from the document metadata and content.
2. Each keyword is converted into an SSE Search Tag using HMAC-SHA256 with a secret backend key, generating an encrypted index of searchable terms.
3. The EHR file is encrypted using AES-256, generating a ciphertext bundle and an AES symmetric key.
4. The AES key is encapsulated using the patient's CRYSTALS-Kyber public key, producing a quantum-safe key bundle that only the patient's Kyber private key can unwrap.

5. The encrypted ciphertext, SSE search tags, and Kyber key bundle are stored in MongoDB. The SHA-256 hash of the file and the upload event are logged as an immutable transaction on the Ethereum blockchain smart contract.

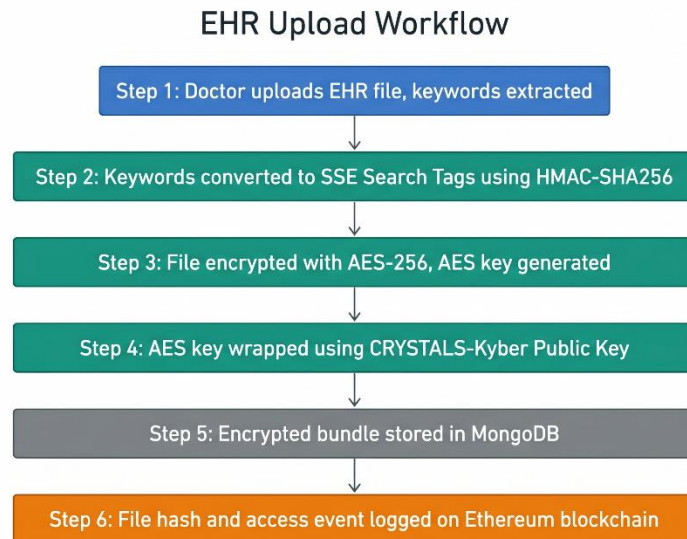


Fig 7.2 EHR Upload Workflow Diagram

7.4 Search and Retrieve Workflow

The search and retrieval process follows a five-step verification pipeline:

1. The authorized user enters a search keyword through the frontend interface. The backend generates a Trapdoor — a keyed HMAC-SHA256 hash of the keyword using the same secret key used during upload.
2. The server matches the Trapdoor against all stored SSE Search Tags in MongoDB. Matching records are identified without the server ever reading or learning the actual keyword or file content.
3. The Ethereum smart contract is queried to verify that the requesting user's MetaMask wallet address is present in the Access Control List for the identified record. If permission is not granted, access is cryptographically denied.

4. The file hash stored on the blockchain is compared against the hash of the retrieved ciphertext to verify data integrity — confirming the file has not been tampered with since upload.
5. The authorized user receives the encrypted bundle. The AES key is unwrapped locally using the patient's Kyber private key, and the file is decrypted on the user's device. The retrieval event is logged as a permanent on-chain transaction.

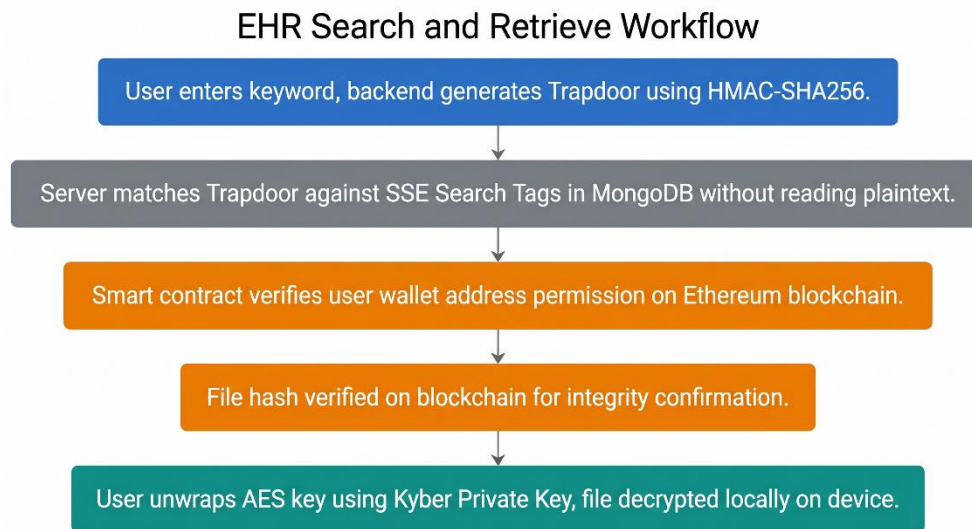


Fig 7.3 Retrieve Workflow Diagram

7.5 Security Guarantees

E-HealthShield provides three mathematically enforced security guarantees:

Quantum Resistance: CRYSTALS-Kyber (ML-KEM, NIST FIPS 203, 2024) ensures that all key encapsulation operations are secure against both classical and quantum adversaries. The system is immune to Shor's Algorithm attacks that would compromise RSA or ECC based systems.

Zero-Knowledge Cloud: The MongoDB storage server operates in a complete zero-knowledge environment. It stores only ciphertext and HMAC-based search tags — it can match

trapdoors against tags and return encrypted records without ever learning the keyword searched, the content of any file, or any patient medical information.

Immutable Audit Trail: Every data access event — upload, search, and retrieval — is permanently recorded as a transaction on the Ethereum blockchain. These records are mathematically immutable and cannot be deleted, modified, or fabricated by any party, providing a cryptographically guaranteed audit trail for complete transparency and accountability.

CHAPTER 8

CONCLUSIONS

8.1 Conclusions

The increasing adoption of Electronic Health Records (EHRs) has significantly improved healthcare management by enabling efficient storage, sharing, and retrieval of patient medical information. However, traditional healthcare data management systems continue to face several challenges related to **data privacy, centralized storage risks, secure accessibility, and transparency in access monitoring**. Furthermore, the emergence of quantum computing introduces additional long-term security concerns for existing cryptographic systems used in healthcare infrastructures.

The proposed project, “**E-HealthShield: A Quantum-Secure and Blockchain-Auditable Framework for Electronic Health Records**,” presents a secure and future-ready approach for protecting Electronic Health Records against emerging cybersecurity threats. The framework integrates **Post-Quantum Cryptography, Searchable Symmetric Encryption, and Blockchain technology** to improve the confidentiality, accessibility, and transparency of healthcare data management systems.

The proposed system utilizes **CRYSTALS-Kyber (ML-KEM)** along with **AES-256 encryption** to ensure secure and quantum-resistant protection of patient medical records. The integration of **Searchable Symmetric Encryption (SSE)** enables efficient keyword-based retrieval of encrypted healthcare data without exposing sensitive patient information. In addition, **Blockchain technology** provides a transparent and tamper-proof audit trail for monitoring healthcare record access and improving accountability.

The proposed E-HealthShield framework aims to provide a **secure, searchable, transparent, and reliable Electronic Health Record system** capable of addressing key limitations of existing healthcare data management approaches. The project also demonstrates the feasibility of integrating modern security technologies into healthcare environments to improve patient privacy and long-term data protection.

In conclusion, **E-HealthShield provides a strong foundation for secure and future-ready Electronic Health Record management systems**, contributing toward safer healthcare data storage and trustworthy digital healthcare infrastructure.

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